



AROR UNIVERSITY
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CSC-331

Analysis of Algorithms

Fall- 2025

Week_04

Lecture Outline

- Asymptotic Notations

Asymptotic Notations

- Asymptotic notations are mathematical tools used to describe the growth rate of an algorithm's time or space complexity as the input size n becomes very large. They give us a way to compare algorithms independent of machine, compiler, or programming style.

Asymptotic Notations

- The **asymptotic notations** we commonly use in algorithm analysis were first **introduced by Donald E. Knuth** (the famous computer scientist and author of *The Art of Computer Programming*).

Asymptotic Notation

- $\Theta, O, \Omega, o, \omega$
- Defined for functions over the natural numbers.
 - Ex: $f(n) = \Theta(n^2)$.
 - Describes how $f(n)$ grows in comparison to n^2 .
- Define a **set** of functions; in practice used to compare two function sizes.
- The notations describe different rate-of-growth relations between the defining function and the defined set of functions.

n input Value

$n \neq 0$ n not

Big O

- Definition:

Ω -notation

Θ -notation

